

The risks posed by AI: Algorithmic discrimination, hallucinations and stochastic parrots, misuse of personal data

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Course Structure

1. Recap & Overview
2. Risks from three levels:
 - Micro: Individuals;
 - Privacy, discrimination, misinformation, manipulation
 - Meso: Enterprises – jurisdictions reactions
 - Macro: Science, Societal and Broader Implications;
 1. Science & Research
 2. Homogenization
 3. Societal Polarization
3. Solutions?
 - Ethical framework
 - Regulation;
 1. Soft (ethics; standard; code of conduct)
 2. Hard
 - Human-in-the-Loop
4. Group Activity



Part I – Recap & Overview of the course

Recap of Last Seminar

Group 3. AI Ethics

Handling of AI beyond what is legally required, beyond the technical aspects, including (but not limited to): social and cultural implications, environmental issues, bias and fairness, etc.

Group 4. AI Hallucination

AI hallucinations are incorrect or misleading results that AI models generate. These errors can be caused by a variety of factors, including insufficient training data, incorrect assumptions made by the model, or biases in the data used to train the model, overfitting with data

Group 5. AI Bias and Fairness

Group 5: AI and Fairness

Here we considered the bias of the company that owns AI in developing or designing the algorithms. For example, some algorithms are designed to reject some applicants for job applications based on maybe the experience, race or whatever bias the company.

Again, the AI company in China called Doubao, when prompted to create an image of persona from different cities in China will create them based on existing stereotypes, making people from Shanghai appear more sophisticated than other lower cities.

It can be inferred that AI cannot be 'fair' in its literal sense as it is not created in vacuum. The fairness of AI is dependent on the developers of the AI or the companies contracting or employing the developers. AI reflects the individual bias or prejudice of people.

Group 6. AI and Anthropomorphism

When AI (non-human) entities is given human-like qualities or behaviour to the point where it is difficult to distinguish from a real human.

eg. humans becoming very dependent on anthropomorphic AI entities for emotional connection, therapy etc...

The background features a grid of light blue squares. A trail of red circles starts from the top right and moves towards the bottom left, following a path that suggests movement or a sequence. The circles are of varying sizes and are placed on the grid squares.

Part II – Risks from Three Levels

Why we talk about AI Risks?

Risks from Micro Level – Privacy is in threat

1. The wall is falling: Privacy and Personal Data in Data Collection & Scraping

- Data Collection and Scraping:

Personal information collection in public domain

Social Media Scraping

- Lack of Informed Consent

Without proper informed consent from individuals. Users may not fully understand how their data will be used or how extensively it is being collected.

Risks from Micro Level – Privacy is in threat

- **Lack of Transparency**

Collection, training, application, deployment behind the scenes

- **Data Overreach**

AI systems sometimes collect **more data than necessary** for their intended purpose.

- **Data Security Risks during Collection**

When large-scale data scraping occurs, especially from **unregulated or non-secure sources**, there are significant security risks and bias involved.

Risks from Micro Level – Privacy is in threat

- **Unfair or Discriminatory Data Collection**

Discriminatory data collection (target specific groups disproportionately)

Biased Data Scraping (predominantly training data on one demographic groups)

- **Use of Sensitive Data Without Safeguards**

Health, financial, or biometric data (e.g., Reddit)

Here's How To Stop X From Using Your Data To Train Its AI

John Koetsier Senior Contributor ©
Journalist, analyst, author, podcaster.

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Jul 26, 2024, 05:02pm EDT

Updated Jul 26, 2024, 05:03pm EDT



Risks from Micro Level - Privacy is in threat

Risks from Micro Level – Algorithmic Shadowing

Processing of Privacy in AI Models – Algorithmic Shadowing

- Even when models are trained using anonymised data, they can still **inadvertently “learn” sensitive information through the patterns** in the data, leading to unintentional privacy violations. The model may encode and perpetuate personal attributes (e.g., gender, race, health conditions).
- Unintentionally “learn” personal details, even though the specific data points may not be explicitly used.
- AI models may retain traces of personal data through their learned parameters, making it difficult to fully separate individual information from the broader model outputs.
- Different parameters means different weight and might be stereotypes, which can lead to discrimination, especially when the model is trained on biased or unrepresentative data.

Risks from Micro Level

AI Output Phase – Jailbreaking, discrimination, disinformation and hallucination

- **AI models could be Jailbroken**

Manipulating or bypassing the restrictions of an AI system

- **Discriminatory outputs**

biased or unfair results based on race, gender, age, or other undefined characteristics

- **Malicious Use:**

Disinformation—false or misleading information deliberately created to deceive others.

Risks from Micro Level

AI Output Phase – Jailbreaking, discrimination, disinformation and hallucination

- **Hallucination:** particularly with large language models (LLMs), hallucination refers to the generation of **factually incorrect or nonsensical information** that **appears to be plausible**, even though it is not **grounded in reality**.
 - **Pattern Recognition:** AI models, especially LLMs, generate text by identifying and continuing patterns in the data they have been trained on. They do not “understand” the information in the way humans do. Instead, they predict what comes next in a sequence based on likelihoods derived from the training data.
 - **No Verification of Facts:** Unlike humans, AI systems do not verify or cross-check the validity of the information they produce. They may combine phrases or concepts that sound convincing but are actually fabricated.
- **Causes:** (i) Insufficient or Ambiguous Training Data; (ii) Over-generalisation (iii) Inference; (iv) Prompts ambiguity
- **Harms:** (i) Misinformation; (ii) Erosion of Trust; (iii) Harmful Decisions; (iv) Reputation damage; (v) Bias Reinforcement

Meso: Enterprises – jurisdictions reactions

- Businesses deploying AI systems may face **discriminatory decision-making**, particularly in hiring, credit scoring, or customer service.
- **Legal Liability and Compliance Risks**: without proper checks and balances face **legal liability**.
 - Compliance risks: GDPR (lawful basis of processing personal data; right to be forgotten);
 - Intellectual property, use copyrighted content
- Over-reliance on AI can lead to **Operational Inefficiencies**, as AI systems might malfunction, make inaccurate predictions, or operate in ways that are not fully understood by human operators.
- AI systems that produce harmful or discriminatory outcomes can lead to a significant **reputational risks** for businesses.
- Trustworthy AI as a selling point!

Meso: jurisdictions reactions

- **Different jurisdictions have conducted difference approaches:**
- **Brussels Effect:**
 - The European Union's approach to AI regulation, exemplified by the AI Act and GDPR (EU Market).
 - Risks: Companies must navigate complex compliance requirements to meet EU standards.
 - Risks related to data infrastructure risks (human rights centric).
- **Washington Reaction:**
 - AI regulation has generally been more market-driven.
 - Favour innovation and economic growth, placing less emphasis on strict AI regulation.
 - Political impact: Trump Effect? Joe Biden's executive order on Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence has been withdrawn by President Trump: potential global AI regulatory divergence; reduced oversight and accountability on AI development.
- **Other jurisdictions?**

Macro: Science, Societal and Broader Implications

Societal Inequalities and Bias:

- Exacerbate wealth and power concentration, deepening social divides.
- Big tech vs. smaller businesses.
- Bias perpetuate discriminatory outcomes in critical sectors.
- AI discrimination marginalises disadvantaged groups (economic disparity and exclusion).

AI in Scientific Research – Scientific Integrity:

- Ethical concerns arise as AI becomes more integral in data analysis and research generation.
- Risks of misinformation and bias in AI-generated research outcomes.
- Loss of accountability in research when AI contributes to data manipulation or inaccurate results.
- Mainstream voice will be amplified, but what about the unconventional research?

Macro: Misinformation, Trust, and Societal Polarisation

Misinformation and Trust:

- Misinformation spread through deepfakes and manipulative content generation.
- AI prioritises sensational or polarising content on social media, amplifying false narratives.
- Erosion of public trust in institutions, media, and information due to AI-driven content manipulation.
- AI-generated misinformation impacts democratic processes, elections, and social cohesion.

Societal Polarisation:

- Echo chambers and pre-existing beliefs and societal divides.
- Sycophancy: Public discourse becomes fragmented
- Difficulty in fostering constructive dialogue across differing ideologies.
- Political manipulation and social engineering exacerbate divisions in society.

Macro: Societal Homogenisation and the Impact of AI on Human Speech

Societal Homogenisation:

- Cultural standardisation: amplifying dominant trends and global preferences.
- Cultural diversity and local expression are marginalised
- Global platforms shape what is seen and heard
- Homogenisation of public experiences.

Human Speech Kicked Off:

- Challenges the authenticity of communication.
- Distortion of public discourse: AI's ability to impersonate individuals.
- AI-driven conversational agents and virtual assistants reshape how people communicate.
- Altering freedom of expression and authenticity in public discussions.

Macro: Group patterns

Group Privacy (e.g., Gene Features):

- AI processes sensitive data like genetic features or biometric data.
- Potential for discrimination based on inherited traits or health predispositions.
- Ethical dilemmas regarding the ownership and use of genetic data in AI research.
- Group privacy concerns arise when AI exploits collective data.



Part III – Solutions

Solutions?

United Nations Educational, Scientific and Cultural Organization (UNESCO)

- First-ever global standard on AI ethics: the ‘Recommendation on the Ethics of Artificial Intelligence’ in November 2021.
- It is applicable to all 194 member states of UNESCO.
- Principles:
 - Proportionality and Do No Harm;
 - Safety and security;
 - Fairness and non-discrimination;
 - Sustainability;
 - Right to Privacy, and Data Protection
 - Human oversight and determination
 - Transparency and explainability
 - Responsibility and accountability
 - Awareness and literacy
 - Multi-stakeholder and adaptive governance and collaboration

Solutions?

Soft Regulation: Ethics, Standards, and Codes of Conduct

Soft regulation focuses on guiding ethical behaviour and establishing voluntary norms that encourage the development of AI technologies in a manner that aligns with societal values. While these frameworks are not legally binding, they serve as guidelines for organisations, industry bodies, and governments to adopt responsible AI practices.

- Ethical Frameworks
- Standards and Best Practices
- Codes of Conduct

Hard Regulation: Legally Binding Laws and Frameworks

While soft regulation sets ethical guidelines and voluntary norms, hard regulation consists of legally binding rules that compel compliance from organisations and impose penalties for non-compliance. These regulations aim to enforce accountability, safeguard fundamental rights, and ensure societal protection against the potential risks of AI.

- The EU AI Act (2021)
- GDPR (General Data Protection Regulation)

Combining Soft and Hard Regulation for Effective AI Governance

Solutions?

Human-in-the-Loop?

Involvement of human oversight or intervention in the decision-making processes of AI systems. It is a critical concept in AI ethics, particularly as AI becomes more autonomous and integrated into various domains such as healthcare, criminal justice, finance, and autonomous vehicles. Human-in-the-loop ensures that AI technologies remain under human control.

Limitations:

- Legal compliance;
- Human value alignment and errors;
- Training and Expertise
- Cost and Resources
- Scalability



Part IV – Group Activity

CREATe Workshop: Revisiting AI Justice in China

This talk will explore the institutional drivers behind China's state led development of AI justice, from Smart Courts to Internet Courts, drawing on empirical evidence. It will also critically assess the promised gains of efficiency and objectivity alongside the legal and constitutional risks raised by these systems. By revisiting the Chinese experience, the lecture offers a timely opportunity to reflect on what AI justice may mean more broadly in the digital age.

Speaker: Prof. Weiwei Shen is visiting professor at Oxford, an Associate Professor at the Law School of China University of Political Science and Law (CUPL), and the Director of the Constitutional Law Department. His research explores the digital transformation of legal frameworks, and he has published widely on themes in law and technology, including constitutional rights in the digital age, platform liability, cyber sovereignty, AI regulation, and online cultural production.

- **Date: 3pm, 28th January**
- **Venue: Room 207, 10 Professor Square**

Space is limited, please send me an email if you would like to attend!!

THANK YOU!

Office hour: Mondays, 2:00-3:00pm.

appointment only

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